

SmartAngan

Offline-First Digital Health Platform for Anganwadi Workers

PILLAR 1 **Smart Measurement System**

BLE Scale + Fallback Guided Entry + Height Methods

PILLAR 2 **Parent Trust & Engagement**

Health Passport + IVR + Physical Cards + WhatsApp

PILLAR 3 **Zero-Tech Inclusion**

No-phone parents + ASHA bridge + Notice board

PILLAR 4 **Decentralisation**

Multi-role + Auto reports + Peer network

PILLAR 5 **Growth Intelligence**

Z-score + Prediction + Family health score

Complete System Overview & Philosophy

1.1 Core Philosophy

SmartAngan is built on one core belief: Make the Anganwadi system so transparent, trustworthy, and easy to use that parents actively compete to enroll their children — not avoid it. Every feature in this system is designed to attack one of three root problems.

Root Problem	Why It Exists	How SmartAngan Solves It
Parents Don't Trust Anganwadi	No visibility into what happens inside the Anganwadi	Health Passports, parent entry SMS, printed cards, trust score
Parents Don't See Value	Free food perception. No awareness of health benefits	Mid-day meal tracking, benefit statements, growth charts
System Depends on One AWW	AWW is absent — everything stops	Audio Diagnostics, register, reports, offline knowledge base

1.2 Complete Architecture — 3 Tiers

TIER 1 — EDGE (AWW's Phone)

- ◆ Flutter Android app — works 100% offline
- ◆ SQLite local database — all data stored on device
- ◆ BLE connection to weighing scale (primary method)
- ◆ Smart guided entry with outlier detection (fallback)
- ◆ Voice entry in Tamil via speech_to_text package
- ◆ Camera-based height measurement — 3 methods built in
- ◆ QR scanner + GPS for tamper-proof attendance
- ◆ On-device WHO Z-score engine — result in 5 seconds

TIER 2 — LOCAL HUB (Raspberry Pi at Centre)

- ◆ Raspberry Pi 4 — local server at Anganwadi
- ◆ Mosquitto MQTT broker — receives sync from AWW phone
- ◆ Local PostgreSQL — aggregates all device data
- ◆ Thermal printer driver — prints monthly health cards
- ◆ Auto-generates weekly notice board report PDF
- ◆ Queues SMS/IVR jobs for when internet available

- ◆ Serves WhatsApp bot for community queries

TIER 3 — CLOUD & OUTREACH

- ◆ Cloud PostgreSQL — receives data from Raspberry Pi
- ◆ CDPO React dashboard — real-time across all centres
- ◆ Parent PWA portal — growth chart via SMS link
- ◆ MSG91 SMS gateway — WhatsApp + regular SMS
- ◆ Exotel IVR — Tamil voice calls for basic phone users
- ◆ Auto supervisor weekly report generation
- ◆ Village nutrition heat map for CDPO

Smart Measurement System — Weight & Height

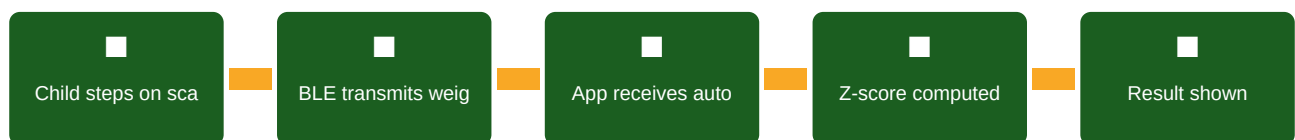
2.1 Weight Measurement — Dual Mode System

SmartAngan uses a **dual-mode weighing system**. BLE scale is the primary method when hardware is available. Smart guided entry is the fallback that works on any existing government-provided scale with zero additional cost.

MODE A — BLE Scale (Primary)

A generic BLE-enabled platform scale (no corporate brand — assembled from open components) connects wirelessly to the AWW's phone. Weight is captured automatically the moment the child steps on the scale.

BLE Flow — Child Mode (2–6 years):



BLE Flow — Infant Mode (0–2 years) — Tare Method:

Same scale, different approach. Mother steps on scale alone first — app records her weight. Mother holds baby and steps on again — app subtracts automatically to get baby's weight. This is the WHO-recommended method for infant weighing.

Formula: Baby Weight = (Mother + Baby reading) - (Mother alone reading)

Used by UNICEF and WHO in field programs. Zero additional hardware. One scale serves all ages.

MODE B — Smart Guided Entry (Fallback — Zero Cost)

When no BLE scale is available, the app transforms the existing government-provided mechanical scale into a smart device through software intelligence alone.

Voice Entry in Tamil

AWW reads the scale aloud. App hears '8 point 3' and auto-fills 8.3 kg. No typing. No transcription error. Works with `speech_to_text` Flutter package offline.

Outlier Detection

App compares new entry against last month's weight. If difference is medically impossible — app flags and asks AWW to re-weigh. Example: 8.2 kg last month, 14.8 kg entered this month — app catches the error.

Range Validation

App knows the WHO possible weight range for every age. Entry outside this range triggers immediate re-check prompt. Eliminates impossible values silently entering the database.

Instant Confirmation

After entry, app shows large number on screen: 'Confirm — is this 8.3 kg?' AWW glances at scale one final time before confirming. Double verification eliminates reading errors completely.

2.2 Height Measurement — Three Methods Built In

The app provides three different height measurement methods. AWW selects based on what is available at her centre. All three work offline, all three are zero additional cost beyond the ₹5 wall chart print for Method 1.

■ METHOD 1 — Wall Chart + Camera OCR

- ◆ Print a height measurement chart on paper — paste on Anganwadi wall permanently
- ◆ Cost: Rs. 5 for printing. One-time. Permanent.
- ◆ Child stands against the chart
- ◆ AWW takes one photo with phone camera
- ◆ ML Kit Text Recognition reads the number from the chart automatically
- ◆ Height recorded — AWW never types anything
- ◆ Accuracy: ± 1 cm. Best accuracy of all three methods.
- ◆ Works on any Android phone. No special hardware.
- ◆ RECOMMENDED METHOD — simplest and most accurate

■ METHOD 2 — AR Measurement via Phone Camera

- ◆ AWW opens AR height screen and points camera at child standing against wall
- ◆ ARCore (built into most Android phones 2018+) detects depth
- ◆ AR overlay shows measurement lines on screen in real time
- ◆ AWW taps the top of child's head on screen
- ◆ App calculates real-world height from depth data
- ◆ Accuracy: ± 1 to 2 cm
- ◆ Flutter package: `ar_flutter_plugin` (free, open source)
- ◆ Works offline. No internet needed.
- ◆ Best for centres without a printed wall chart

■ METHOD 3 — ML Kit Pose Detection

- ◆ AWW points camera at child standing straight against any wall
- ◆ Google ML Kit Pose Detection API detects 33 body landmarks
- ◆ App identifies top of head and bottom of feet landmarks
- ◆ Calculates height using pixel ratio and known reference
- ◆ Accuracy: +2 to 3 cm (acceptable for growth monitoring)
- ◆ Works completely offline — ML Kit runs on device
- ◆ Free. No API key. No internet.
- ◆ Best fallback when no wall chart and older phone without ARCore

Method Selection Logic in App:

Situation	Recommended Method	Why
Wall chart printed and available	Method 1 — OCR	Most accurate, simplest
No chart, modern phone (2018+)	Method 2 — AR	Good accuracy, no print needed
No chart, older phone	Method 3 — Pose Detection	Works on any Android
Child won't stand still (toddler)	Any + gamified mode	Animation makes child stand still

2.3 Gamified Measurement for Toddlers

Children aged 2–4 often refuse to stand still for measurement. The app includes a gamified mode that makes it fun:

- ◆ App shows animated character: 'Stand like a soldier! Can you be as tall as this hero?'
- ◆ Child naturally stands straight trying to match the animation
- ◆ Height measurement captured during this still moment
- ◆ Child gets a virtual star reward on screen after measurement
- ◆ Parents watching smile — positive association with Anganwadi visits built instantly

3.1 The Parent POV Problem

Parent Type	% of Rural TN	Current Experience	SmartAngan Experience
Smartphone + WhatsApp	20%	No updates from Anganwadi	Health Passport + Milestone messages
Basic phone (calls/SMS only)	45%	No contact at all	Tamil IVR voice call after each visit
No phone	15%	Zero information	Physical health card + ASHA visit
Shared family phone	8%	Occasional SMS	WhatsApp bot — query anytime
Elderly guardian	2%	No tech access	Physical card + panchayat announcement

3.2 Child Health Passport (Smartphone Parents)

After every measurement, parent receives a WhatsApp message with a link. No app to download. No login. Just tap and see.

- ◆ Child's name, age, latest weight and height
- ◆ WHO growth chart plotted over time — visual and easy to understand
- ◆ Nutritional status — Normal / MAM / SAM with color indicator
- ◆ Vaccination completion percentage and next due date
- ◆ Attendance this month
- ◆ AWW contact number
- ◆ Link is tokenized and expires in 7 days — secure

3.3 Tamil IVR Voice Call (Basic Phone Parents)

For parents with basic phones — no WhatsApp, no smartphone — an automated voice call in Tamil is triggered after each Anganwadi visit.

Sample call script (Tamil):

"Vanakkam. Ungal pillai Priya indru Anganwadiyil irundhaal. Uzhal: 9.2 kilo. Nilai: Nallirukku. Maarkalanilai kooppu: April 15. Nandri."

- ◆ Works on any phone including 10-year-old Nokia feature phones
- ◆ Tamil language — no literacy required to understand
- ◆ Parent hears child's name — immediate emotional connection
- ◆ Cost: Rs. 0.30 to 0.50 per call via Exotel or Ozonetel (Indian IVR providers)
- ◆ Call triggered automatically when AWW saves measurement

- ◆ No action needed from AWW — fully automated

3.4 Physical Health Card (No Phone Parents)

Every month, when measurements are recorded, SmartAngan automatically generates a printable health card. AWW prints it on a basic thermal printer (Rs. 3,000 one-time) and hands it to the parent at pickup time.

- ◆ Child's name, age, date
- ◆ Weight and height with Normal/MAM/SAM status
- ◆ Next vaccination due — vaccine name and date
- ◆ Next Anganwadi visit date
- ◆ AWW phone number
- ◆ Simple icons for illiterate parents — green face for Normal, yellow for MAM, red for SAM
- ◆ Cost: Rs. 2 to 3 per card (thermal paper)
- ◆ Parent keeps it like a school report card — shows family members

3.5 Milestone Celebration Messages

When a child hits a health milestone, parent receives a celebration WhatsApp message. This is the viral enrollment growth mechanism — mothers share these in family groups.

- ◆ Trigger: Completed all 6-week vaccinations successfully → Auto WhatsApp celebration message sent
- ◆ Trigger: Weight moved from MAM to Normal range → Auto WhatsApp celebration message sent
- ◆ Trigger: Attended Anganwadi 20+ days this month → Auto WhatsApp celebration message sent
- ◆ Trigger: Completed 6 months of exclusive breastfeeding → Auto WhatsApp celebration message sent
- ◆ Trigger: First birthday health report — full growth summary → Auto WhatsApp celebration message sent

3.6 Anganwadi Trust Score

Each Anganwadi centre gets a publicly visible trust score based on measurable outcomes. Parents can compare centres. High-scoring centres attract more enrollment naturally.

Score Component	Weight	How Measured
Attendance rate this month	25%	Records in app
Vaccination coverage	25%	Vaccination module
Daily growth monitoring	20%	Frequency of measurements
Parent satisfaction rating	20%	1-5 star rating via IVR or WhatsApp
SAM referral follow-up rate	10%	Referral tracking module

4.1 Multi-Role Login System

Role	Access Level	Use Case
Primary AWW	Full access	Day-to-day operations
Helper AWW / Anganwadi Assistant	Dist center only	Assists primary AWW
ASHA Worker	Home visit module only	Tracks home visits and referrals
Substitute AWW	Temporary full access	Covers when primary is absent
ICDS Supervisor	View + alerts only	Remote monitoring
CDPO	Full dashboard + reports	Block-level oversight

If primary AWW is absent: Substitute logs in. Same data. Same children. Zero interruption. Supervisor is automatically notified of substitution.

4.2 Offline AWW Knowledge Base

Built into the app — an offline reference guide that removes dependency on the supervisor for routine decisions:

- ◆ What to do when a child is classified SAM — step-by-step referral protocol
- ◆ Nutrition counseling scripts for different age groups in Tamil
- ◆ How to handle a child who refuses weighing — child psychology tips
- ◆ Danger signs in pregnant mothers — when to refer immediately
- ◆ Government scheme eligibility and enrollment process
- ◆ Emergency contacts — PHC, NRC, ICDS supervisor, CDPO

4.3 Automated Daily Tasks — System Runs Without AWW Action

Time	Automated Task	Triggered By
Every morning	Attendance reminder SMS to parents	Scheduler on Raspberry Pi
After each measurement	Weight update to parent (IVR/SMS/WhatsApp)	Measurement save event
Every evening	Daily summary to supervisor	End-of-day scheduler

Time	Automated Task	Triggered By
2 days before vaccine due	Vaccination reminder to parent	Vaccination scheduler
Every Monday	Weekly block report to CDPO	Weekly scheduler
1st of every month	Monthly health card generation for all children	Monthly scheduler
3 missed days	Absence alert to AWW and supervisor	Attendance checker

4.4 Village Notice Board — Auto Printed Weekly Report

Every week, SmartAngan generates a one-page printed report for the village notice board. AWW prints it and pins it up. Zero technology required for the community to read it.

- ◆ Children monitored this week and nutrition status summary
- ◆ Vaccinations completed this week
- ◆ Meals served — days and types
- ◆ AWW on duty name and contact
- ◆ Next health check date and time
- ◆ Enrollment appeal — 'Your child can join. It is free.'
- ◆ Cost: Rs. 2 for printing. Auto-generated — AWW just presses print.

5.1 Morning Workflow — Child Arrives at Anganwadi

7:30 AM	AWW Opens App	Logs in. App loads offline from SQLite. Today's child list shown.
8:00 AM	Children Arrive	AWW scans each child's QR card at entry. GPS auto-captured.
8:15 AM	Measurement Time	BLE scale primary. If unavailable, voice guided entry fallback.
8:16 AM	Height Measured	AWW selects method — OCR chart, AR, or pose detection.
8:17 AM	Z-Score Result	App shows Normal/MAM/SAM instantly. SAM triggers referral alert.
8:18 AM	Parent Notified	SMS/IVR/WhatsApp sent to parent automatically in background.
9:00 AM	Sync to Pi	App connects to Raspberry Pi WiFi. MQTT sync — all data uploaded.
Evening	Pi Syncs to Cloud	When internet available — cloud updated. Dashboard refreshed.

5.2 New Child Enrollment Workflow

- ◆ AWW registers new birth — enters name, DOB, gender, mother phone
- ◆ App auto-creates child profile in SQLite
- ◆ Vaccination schedule auto-calculated from date of birth
- ◆ QR code generated for the child — printed for family
- ◆ Child added to AWW's daily list, supervisor's dashboard, all modules
- ◆ Welcome SMS/IVR sent to mother — enrollment confirmed
- ◆ If mother has no phone — physical enrollment card printed
- ◆ First measurement scheduled — AWW notified for home visit

5.3 SAM Detection & Referral Workflow

DETECTION: Weight entered or BLE captured → Z-score computed → Below -3 → SAM flag

IMMEDIATE ALERT: Red screen on AWW phone — 'SEVERE MALNUTRITION — REFER IMMEDIATELY'

REFERRAL PROTOCOL: App shows step-by-step: Contact PHC, fill referral form, note NRC contact

SUPERVISOR ALERT: Push notification sent to ICDS supervisor automatically

PARENT NOTIFICATION: IVR call or WhatsApp — 'Please meet AWW urgently for your child'

FOLLOW-UP TRACKING: App tracks whether referral was acted on. Reminder if pending after 48 hrs.

CDPO VISIBILITY: SAM case appears on CDPO dashboard within minutes of detection

RESOLUTION: AWW marks referral complete. NRC outcome recorded. Growth monitored closely.

Development Roadmap — Review by Review

REVIEW 0 — FOUNDATION | Months 1-2

Goal: Working core app. Child registration + Z-score + BLE + guided entry fallback.

What to Build:

- ◆ Flutter project setup with complete folder structure
- ◆ SQLite schema — all 8 tables created and tested
- ◆ Child registration and list screen
- ◆ WHO Z-score engine — LMS formula with bundled data tables
- ◆ BLE scale integration — flutter_blue_plus package
- ◆ Tare method for infant weighing in BLE mode
- ◆ Smart guided entry — voice input + outlier detection + range validation
- ◆ Mode selector — BLE primary, guided fallback
- ◆ Basic growth recording screen with live Z-score result
- ◆ SAM alert screen with referral protocol

Demo: Register child → BLE weight captured → Z-score shown in 5 seconds → SAM alert triggered

REVIEW 1 — FULL FEATURES | Months 3-4

Goal: All 7 core features working. Height measurement. Attendance. Maternal. Vaccination.

What to Build:

- ◆ Wall chart OCR height measurement — Method 1
- ◆ AR height measurement — Method 2 (ar_flutter_plugin)
- ◆ ML Kit Pose Detection height — Method 3
- ◆ Gamified measurement mode for toddlers
- ◆ QR code generation for all beneficiaries
- ◆ QR scanner + GPS attendance — geolocator package
- ◆ Maternal health tracking module — BP, HB, weight, ANC
- ◆ Maternal risk flagging — automated danger sign alerts
- ◆ Vaccination schedule auto-calculation from DOB
- ◆ Auto-enrollment trigger on birth registration
- ◆ Multi-role login system — AWW, helper, ASHA, supervisor
- ◆ Raspberry Pi setup — MQTT broker running and tested
- ◆ MSG91 SMS integration — test messages sent
- ◆ Tamil IVR integration — Exotel test call

Demo: Complete AWW day: birth registered → enrolled → weighed (BLE+height) → QR attendance → maternal visit → sync to Pi

REVIEW 2 — INTEGRATION & OUTREACH | Months 5-6

Goal: Cloud sync working. All parent outreach channels live. Field visit completed.

What to Build:

- ◆ Cloud PostgreSQL deployed — Pi syncing to cloud
- ◆ Parent PWA portal — growth chart, vaccination, full UI
- ◆ Health Passport via WhatsApp link
- ◆ Thermal printer integration — monthly health card auto-print
- ◆ Weekly notice board report auto-generation and print
- ◆ WhatsApp community bot for parent queries
- ◆ CDPO React dashboard — all 7 modules
- ◆ Village nutrition heat map
- ◆ Offline AWW knowledge base — built into app
- ◆ Auto supervisor weekly report
- ◆ Trust score calculation and display
- ◆ Anganwadi field visit — document AWW feedback
- ◆ End-to-end testing on low-end Android phones
- ◆ Security audit — tokenized links, encrypted sync

Demo: Full system live: BLE → Z-score → sync → cloud → dashboard → SMS → IVR → parent card printed

REGIONAL REVIEW — FINAL PRODUCT | Month 7+

Goal: Pilot in 2-3 Tamil Nadu Anganwadis. Real data. Real AWW feedback. Real impact metrics.

What to Build:

- ◆ Pilot deployment in Coimbatore area Anganwadis
- ◆ Real child and maternal data collected
- ◆ AWW training — 2-3 hours per centre
- ◆ Measure: paperwork time reduced, SAM detection speed, AWW satisfaction
- ◆ Measure: parent enrollment rate before and after
- ◆ Measure: vaccination coverage improvement
- ◆ Document AWW quotes and field observations
- ◆ Growth trend prediction feature (AI angle — linear regression)
- ◆ Family health score module
- ◆ Sibling linking feature
- ◆ Seasonal malnutrition prediction using historical data
- ◆ Pilot report with before/after metrics

◆ Business model finalization — B2G pricing

Demo: *Present pilot data. Show before/after. Let panel interact with live system. AWW testimonial video.*

Field Visit Plan & Complete Innovation Summary

7.1 Anganwadi Field Visit — Must Do Before Review 1

A field visit is not optional bro. When you stand in front of a panel and say 'I visited the Peelamedu Anganwadi and the AWW showed me 8 registers' — your entire credibility multiplies. Here is exactly what to do.

How to Find a Centre Near Coimbatore:

- ◆ Google 'Anganwadi centre Peelamedu' or 'Anganwadi Kinathukadavu'
- ◆ Visit the CDPO office — explain you are a student doing a project
- ◆ Best time to visit: 9am-11am any weekday
- ◆ Take your phone for photos, a notebook, and this question list

Questions to Ask the AWW:

- ◆ How many registers do you maintain? Can you show me each one?
- ◆ How do you currently weigh children? What type of scale?
- ◆ How long does it take to fill all registers per day?
- ◆ What happens when you make an error in the register?
- ◆ Do you use POSHAN Tracker? Why or why not?
- ◆ Do you have a smartphone? What kind?
- ◆ How do you inform parents about their child's health?
- ◆ What is your biggest challenge every day?
- ◆ If you had an app that did all this automatically — would you use it?
- ◆ What would make an app easy for you to use?

7.2 Complete Innovation Summary — One Table

Innovation	Pillar	Cost	Impact	Review
BLE Scale — primary weight	Measurement	Hardware	Eliminates errors	R0
Tare method — infant weight	Measurement	Rs. 0	Serves 0-2 yr infants	R0
Smart guided + voice entry	Measurement	Rs. 0	Fallback for all centres	R0
Wall chart OCR — height	Measurement	Rs. 5	Best accuracy height	R1
AR height measurement	Measurement	Rs. 0	Modern phone option	R1
ML Kit pose — height	Measurement	Rs. 0	Any phone fallback	R1

Innovation	Pillar	Cost	Impact	Review
Gamified toddler mode	Measurement	Rs. 0	Solves cooperation issue	R1
Child Health Passport PWA	Parent Trust	Rs. 0.15/SMS	Smartphone parents	R1
Tamil IVR voice call	Parent Trust	Rs. 0.40/call	Basic phone parents	R2
Physical health card print	Parent Trust	Rs. 3/card	No-phone parents	R2
Milestone WhatsApp messages	Enrollment	Rs. 0.15	Viral word of mouth	R2
Anganwadi trust score	Enrollment	Rs. 0	Creates healthy competition	R2
Village notice board report	Community	Rs. 2/week	Zero-tech community view	R2
WhatsApp community bot	Community	Rs. 0	Any parent can query	R2
Multi-role login	Decentralise	Rs. 0	No single point failure	R1
AWW knowledge base	Decentralise	Rs. 0	Reduces supervisor calls	R1
Auto supervisor report	Decentralise	Rs. 0	Efficient supervision	R2
Absence alert system	Decentralise	Rs. 0	No child missed	R2
Village nutrition heat map	Intelligence	Rs. 0	CDPO targeting	R2
Growth trend prediction	Intelligence	Rs. 0	AI angle for panel	Regional
Family health score	Intelligence	Rs. 0	Holistic family view	Regional
Seasonal malnutrition alert	Intelligence	Rs. 0	Predictive not reactive	Regional

■ One Line To Tell Your Teacher

'SmartAngan is designed to attack child malnutrition from three angles simultaneously — by making measurement accurate and effortless through BLE scale with smart guided fallback, by making the Anganwadi system visible and trustworthy to every parent regardless of whether they have a smartphone, basic phone, or no phone at all through Health Passport, Tamil IVR calls, physical printed cards, and community notice boards, and by removing single-point dependency through multi-role login, automated daily tasks, and an offline AWW knowledge base — so that every child in every village gets the nutrition monitoring they deserve, delivered by a system that works even when the internet doesn't.'